

Ex-Ante ERW — Results dashboard

Profitability maps and tables under different assumptions

2026-05-26

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Profitability under different assumptions

This is a focused results dashboard. It complements `erw-model.md` (full methodology) and `erw-tutorial.md` (worked example). All figures and tables below are generated from the 276 gross-margin rasters under `data/economics_erw/` by `erw/erw-10-profitability-maps-tables.R`.

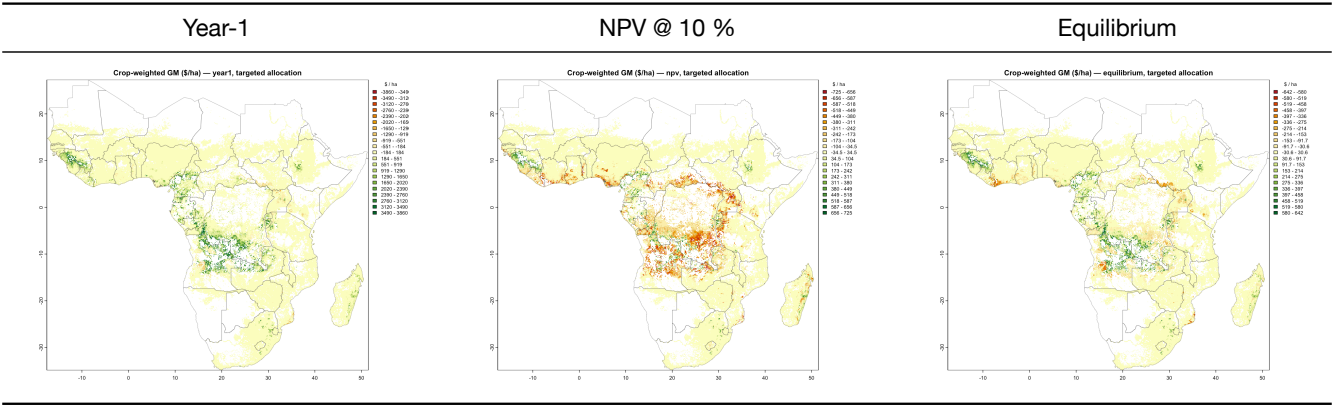
Scope: Sub-Saharan Africa, 23 SPAM crops, 0.083° working grid. **Regimes:** year-1 (one application, agronomic + CDR revenue in year 1) · NPV @ 10 % (CDR revenue phased over 5 years, agronomic returns over the reapplication interval) · equilibrium (steady-state maintenance dose). **Allocations:** targeted (LiTAS dose only on acid pixels above each crop's tolerance) · `uniform_10` / `uniform_20` / `uniform_50` (uniform t basalt / ha across all cropland regardless of soil acidity). **Default parameters (refreshed 2026-05-26):** carbon price \$150 / tCO₂, MRV **\$20 / tCO₂** (lowered from \$30), **grain size 50 µm** (lowered from 100 µm; the Streffer grinding-curve anchor), \$10 / t quarry gate, country-level electricity tariffs and grid CI, spatial delivered-transport costs, year-1 agronomic return **summed over the 5-year basalt residency** (was a single-season figure), CGIAR-CSI Global Aridity Index for the pedogenic deduction, per-crop EcoCrop max-pH cliffs (range 5.5–8.0).

1. Continent-wide gross margin

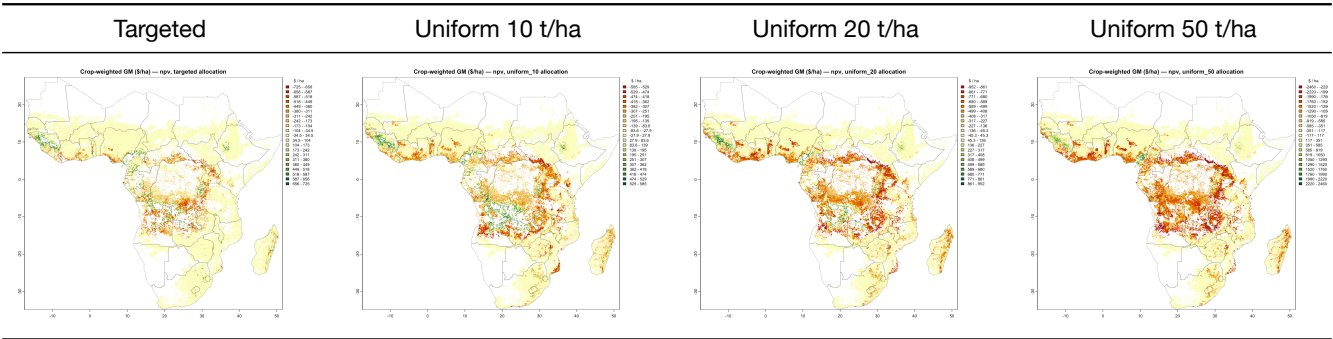
Each map below shows the crop-weighted average gross margin per hectare (\$ / ha) at the 0.083° pixel — weighted by harvested area across the 23 crops sharing that pixel. Green pixels are profitable; red

are loss-making.

1.1 By return regime (LiTAS-targeted allocation)



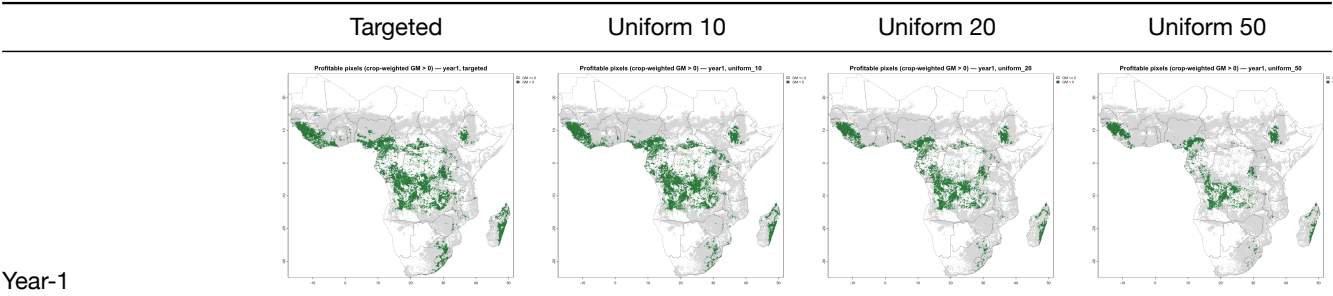
1.2 By allocation rule (NPV regime)

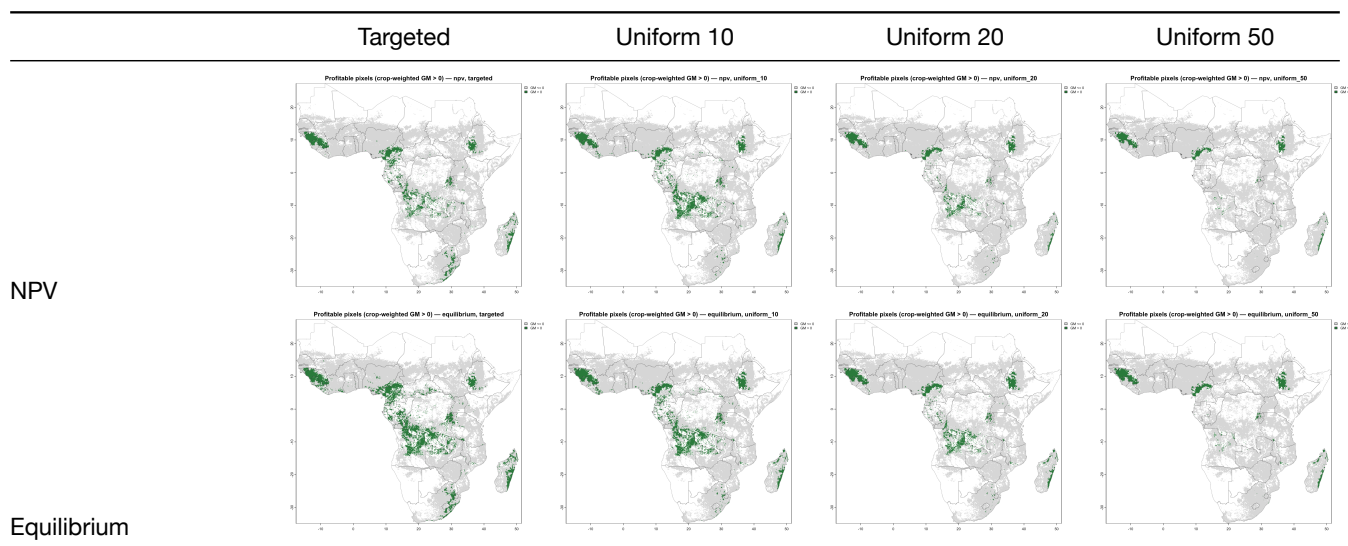


The targeted allocation prunes the action set to acidic pixels and pays the LiTAS-calibrated dose; uniform rules pay a fixed rate across all cropland and recover CDR even on near-neutral soils that the targeted rule skips.

2. Profitable footprint

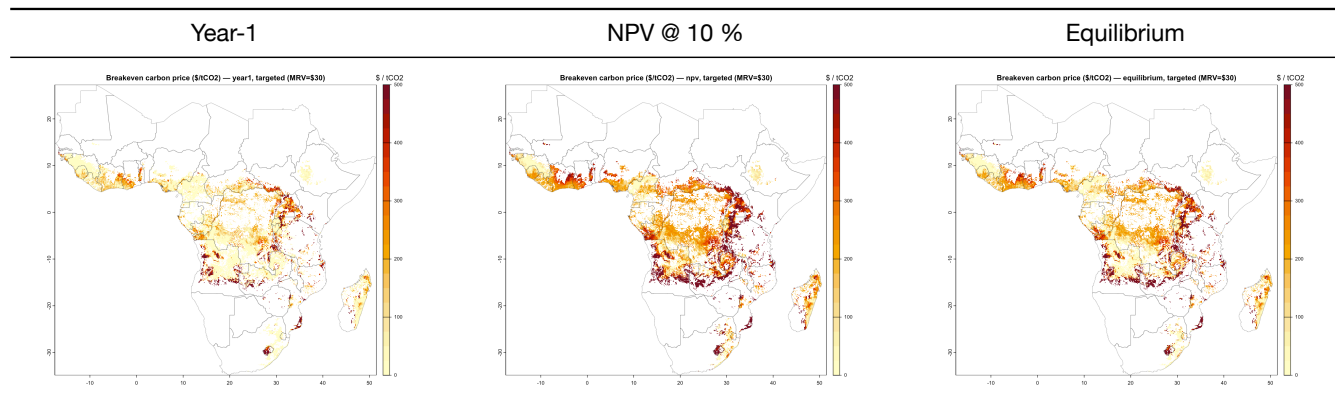
Binary masks showing where the crop-weighted GM is strictly positive. These are the candidate pixels for the supply-ranked deployment in erw-supply.





3. Breakeven carbon price

For each pixel, the minimum carbon price (\$ / tCO₂) that drives crop-weighted GM to zero, holding all other defaults constant (MRV \$20 / tCO₂, grain size 50 µm, etc.). Pixels with no CDR potential are masked; the legend caps at \$500 / tCO₂ for legibility.



The equilibrium regime tends to need higher carbon prices because the maintenance dose recurs forever, while year-1 / NPV amortise the up-front agronomic uplift against a single application.

4. Per-crop gross margin (selected crops, targeted)

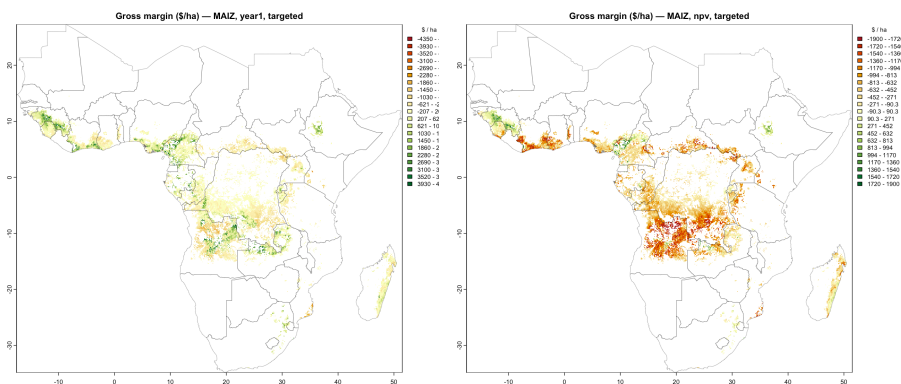
The five highest-acreage crops, year-1 and NPV regimes:

Crop

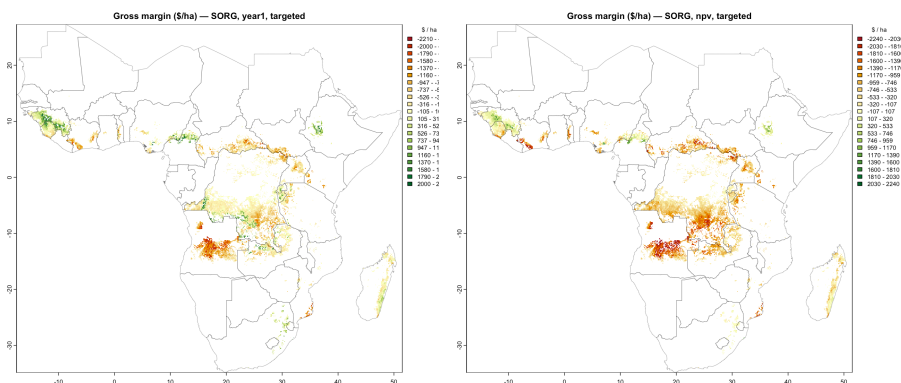
Year-1

NPV @ 10 %

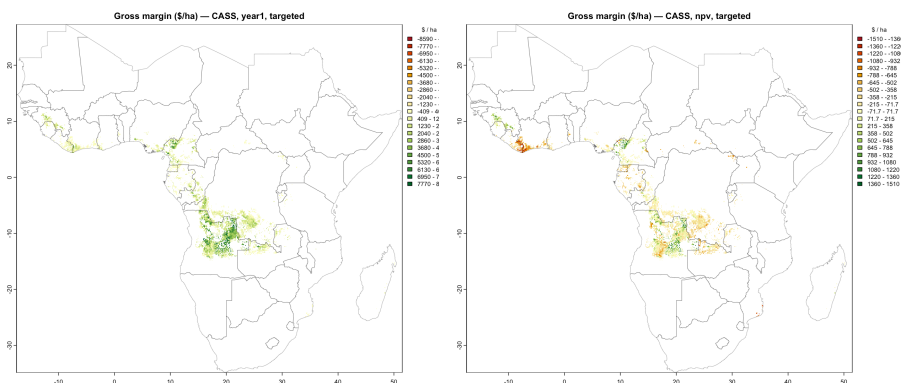
Maize (MAIZ)



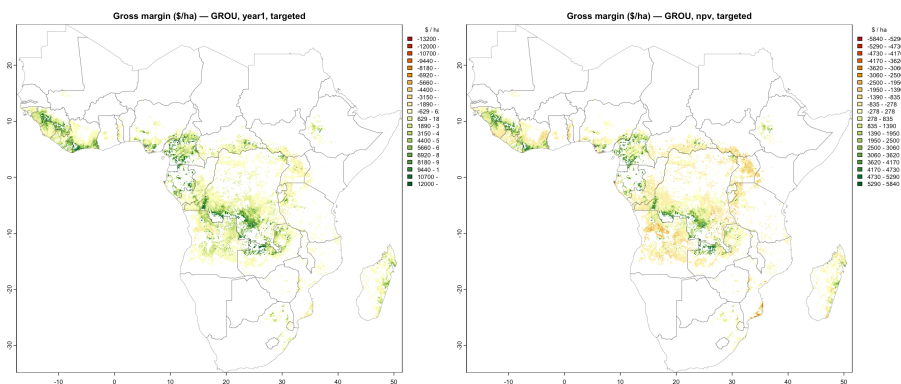
Sorghum (SORG)



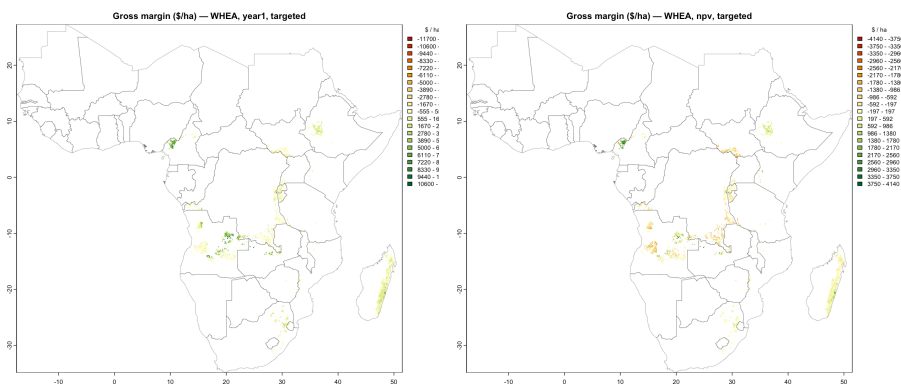
Cassava (CASS)



Groundnut (GROU)



Wheat (WHEA)



5. SSA-wide totals — one row per regime × allocation

Aggregates are summed across all 23 crops, all pixels (no positive-GM filter unless the column says “profitable”). Source: docs/tables/output-ssa-summary.csv.

Regime	Allocation	Total GM (M\$)	Agro return (M\$)	Basalt cost (M\$)	CDR revenue (M\$)	Total CDR (Mt)	Profitable area (Mha)	CDR on profitable (Mt)
year1	targeted	18439.6	30219.9	21545.4	10565.7	81.27	137.87	58.41
npv	targeted	-1106.7	12029.7	21545.4	8390.7	81.27	132.41	37.43
equilibrium	targeted	1965.3	6044	7363.9	3442	26.48	134.88	13.6
year1	uniform_10	19950.3	30219.9	75918.2	13742.4	105.71	127.07	81.59
npv	uniform_10	-6106.2	5494.5	75918.2	10913.5	105.71	119.24	68.31
equilibrium	uniform_10	-4200.4	6044	75918.2	13742.4	105.71	120.61	71.02
year1	uniform_20	9712.2	30219.9	151836.4	27484.8	211.42	124.08	153.67
npv	uniform_20	-17701.2	5494.5	151836.4	21827	211.42	117.41	130.4
equilibrium	uniform_20	-14438.5	6044	151836.4	27484.8	211.42	118.64	135.36
year1	uniform_50	-21002	30219.9	379591	68712.1	528.55	120.61	355.11
npv	uniform_50	-52486.3	5494.5	379591	54567.4	528.55	116.07	314.58
equilibrium	uniform_50	-45152.7	6044	379591	68712.1	528.55	117.09	325.44

6. Top crops by aggregate gross margin (NPV regime)

The full crop × regime × allocation matrix is in docs/tables/output-crop-ranking.csv. Below: the top eight crops by total GM under the NPV-targeted scenario.

Rank	Crop	Total GM (M\$)	Profitable area (Mha)	Total CDR (Mt)
1	BEAN	1285.63	4.749	17.142
2	GROU	874.78	10.537	11.838
3	POTA	872.05	1.408	2.793
4	SWPO	48.11	2.963	4.925
5	LENT	20.98	0.103	0.068
6	WHEA	14.07	2.904	0.687
7	BARL	6.45	1.212	0.256

Rank	Crop	Total GM (M\$)	Profitable area (Mha)	Total CDR (Mt)
8	CHIC	5.2	0.41	0.114

7. Country breakdown — NPV, targeted (top 12 by GM)

Full per-country table for all regime × allocation combinations: docs/tables/output-country-summary.csv (528 rows = 44 countries × 12 scenarios). Below: the top dozen by aggregate GM in the NPV-targeted scenario.

Rank	ISO3	GM (M\$)	Profitable area (Mha)	Total CDR (Mt)
1	CMR	3313.11	0.843	17.56
2	GIN	567.24	0.561	3.974
3	RWA	369.48	0.418	3.121
4	ETH	297.15	0.654	5.271
5	BDI	154.63	0.275	2.109
6	SLE	41.2	0.112	1.526
7	ZAF	24.08	0.072	0.051
8	GAB	12.61	0.017	0.411
9	SWZ	0.59	0.005	0.024
10	GNQ	-1.17	0.009	0.196
11	MDG	-1.27	0.1	0.813
12	SEN	-1.56	0	0.003

8. How to rebuild

1. Make sure erw-7 outputs exist

```
ls data/economics_erw/targeted/ # should have 69 = 23 crops × 3 regimes
```

2. Render maps + tables

```
Rscript erw/erw-10-profitability-maps-tables.R
```

3. Rebuild this PDF

```
bash docs/build-erw-results.sh
```

The R script is idempotent and ~3 min on an M-series laptop. The PDF build requires pandoc + xelatex (same toolchain as erw-model.pdf).